

Introduction to Statistics for HCI Using Jamovi

A course for HCI practitioners and researchers on inferential statistics and Jamovi, an open-source statistics software, comparable to IBM SPSS

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ABSTRACT

When developing and improving products in a human-centered way or testing a hypothesis in your HCI research, you will very likely want to collect some quantitative data like user satisfaction or user performance and make sense of it. Is your design faster than the competitor? Does this new technology really lead to fewer errors?

But alas, testing hypotheses... don't you need to understand statistics for that? And isn't statistics just boring mathematics, but harder?

Do you want to learn how to do basic statistical tests on HCI-related data with a free open-source tool and have little to no affection for math? Then this course is for you!

CCS CONCEPTS

• HCI design and evaluation methods; • Empirical studies in HCI; • Hypothesis testing and confidence interval computation;

KEYWORDS

Inferential Statistics, Statistics Software, User Testing, Usability Testing

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1 BENEFITS

This course will provide attendees with basic knowledge about statistics for planning and conducting common usability and UX evaluation experiments that measure quantitative data. The course gives an introduction to inferential statistics, which is needed to analyze and interpret such quantitative data. It will focus the practical application to be accessible for all by omitting strong requirements of mathematical knowledge. Besides the theoretical background,

the course will introduce the attendees to Jamovi, a free and open-source statistical software, which can replace commercial programs like IBM SPSS for most HCI-related forms of data analysis. They will acquire practical skills in applying Jamovi to exercise questions with prepared datasets, which should make the transfer to real-world problems easy. Anyone new to collecting and analyzing quantitative data will benefit from this course.

2 INTENDED AUDIENCE(S)

People that plan to analyze quantitative HCI data but have little or no prior knowledge of inferential statistics and hypothesis testing. People that have basic stats knowledge but want to get to know a free open-source tool to do the heavy lifting for them. Students, academic beginners, practitioners.

3 PREREQUISITES

High-school-level math knowledge. No statistics knowledge needed. Knowledge of user testing and the Human-Centered Design Process is beneficial, but not required.

4 CONTENT

The course consists of a theoretical and a practical part. The first part will consist of very short explanations of statistical concepts based on examples relevant to HCI research alternating with Q&A. It will cover the following topics:

- why inferential statistics? difference to descriptive statistics
- types of studies: experiment, quasi-experiment, correlation study
- hypotheses: why hypotheses, what is the null hypothesis and why is it useful, one-sided vs. two-sided
- variables: what are independent and dependent variables
- scale of measure: what are nominal, ordinal, interval and ratio data and why is it important
- effect size: what is it, what are typical effect sizes in HCI research and why
- errors: what are type I and type II errors
- significance level: why is it needed and how is it decided
- population and sample: how to pick a sample, limits of sampling in HCI research, sample size
- t-Test: what is a t-Test and why is it useful
- analysis of variance: what is an ANOVA and why is it useful, what are post-hoc tests and why are they needed, accumulation of errors

Attendees will receive a hand-out with explanations of those concepts and their relevance to HCI research afterwards.

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The second part of the course will go over the user interface of Jamovi:

- creating and editing variables
- importing and editing data
- running analyses: t-test and ANOVA
- editing reports
- exporting R code

After the introduction to Jamovi, attendees will work on exercises trying to solve realistic research questions.

5 PRACTICAL WORK

The second part of the course will be a hands-on introduction to the Jamovi user interface. Then, different HCI-related datasets will be analyzed using the software. Some research questions with realistic datasets will be provided. Attendees are encouraged to bring their own computers to conduct analyses in groups under the supervision of the instructor.

6 INSTRUCTOR BACKGROUND

Jurek Breuninger is a professor of User Experience Design at IU International University. He is responsible for “Introduction to Usability Evaluation” in German and “Introduction to User Testing” in English besides other HCI-related courses. Before that, he was a Usability/UX Researcher for twelve years in academia and business, mainly touchscreen interaction, automotive HMI and e-commerce. He has given trainings in experiment design and statistics to practitioners in different domains. His dissertation largely consists of statistical analysis of quantitative data from user experiments with

software. He also gives introductions to Jamovi at local UX Research meetups.

7 RESOURCES

The course draws content from the author’s lecture notes and textbooks on usability/UX testing and statistics like those in the references, but is very condensed for the format.

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